Module 20

Entity-Relationship Model/3

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Module Recap

- ER Diagram for ER Models

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- ER Diagram for ER Models
- Translation of ER Models to Relational Schema

Module Objectives

Goals

- To understand extended features of ER Model

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- To understand extended features of ER Model
- To discuss various design issues

Module Outline

- Extended ER Features

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- Design Issues

Extended ER Features

Non-binary Relationship Sets

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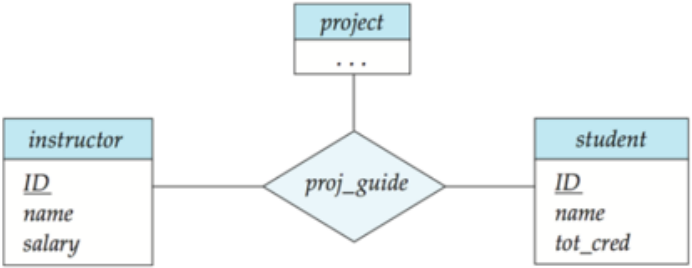

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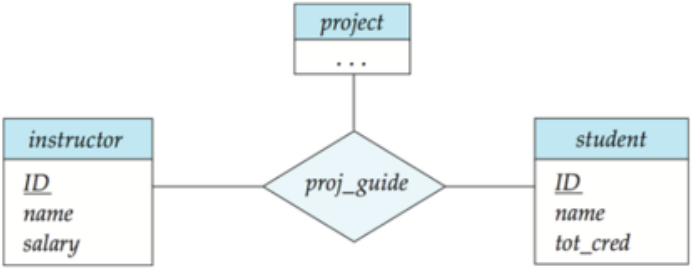

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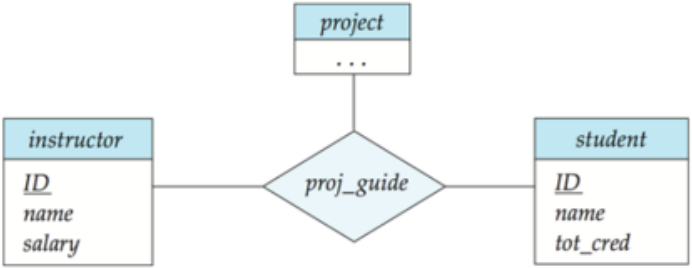
- ER Diagram with a Ternary Relationship



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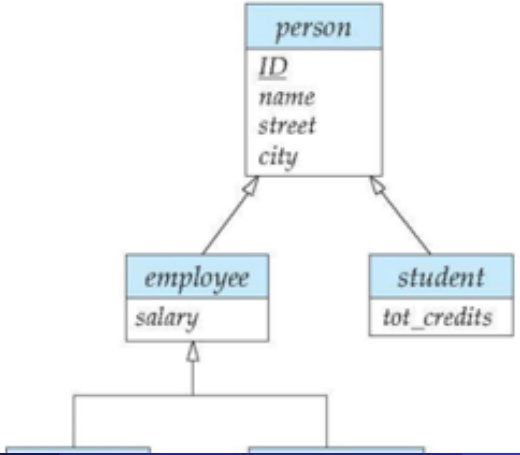
- **Attribute inheritance**: A lower-level entity set inherits all the attributes and relationship participation of the higher-level entity set to which it is linked

Specialization: ISA (2)

- Overlapping: employee and student



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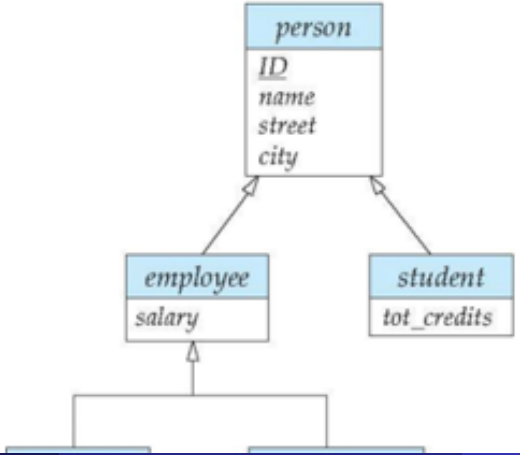

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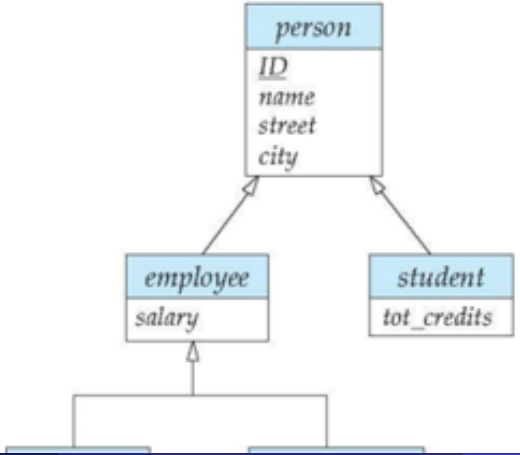

Representing Specialization via Schema

- **Method 1:**

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- **Drawback**: Getting information about, an employee requires accessing two relations, the one corresponding to the low-level schema and the one corresponding to the high-level schema

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Representing Specialization as Schema (2)

- **Method 2:**
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- **Drawback**: name, street and city may be stored redundantly for people who are both students and employees

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Design Constraints on a Specialization / Generalization

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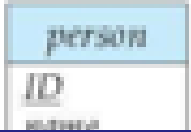

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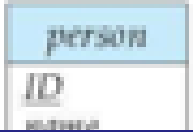

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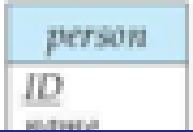

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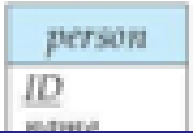

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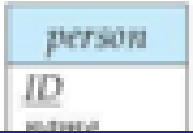
- The `student` generalization is total. All student entities must be either graduate or undergraduate. Because the higher-level entity set arrived at through generalization is generally composed of only those entities in the lower-level entity sets, the completeness constraint for a generalized higher-level entity set is usually total.



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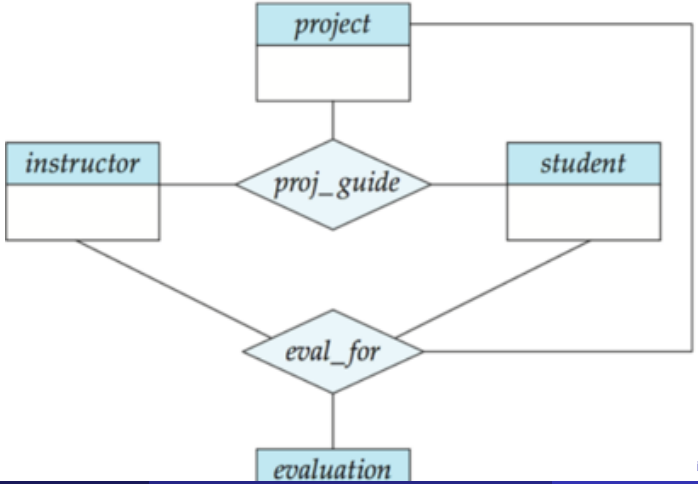

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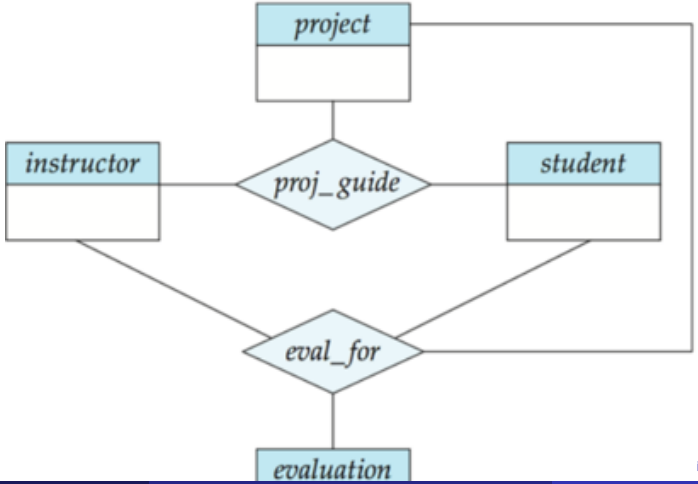

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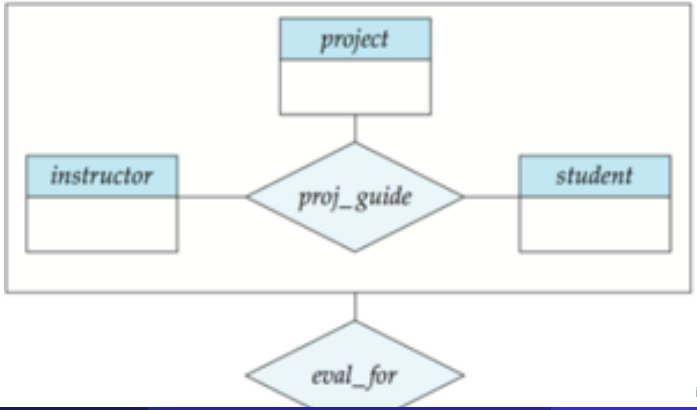

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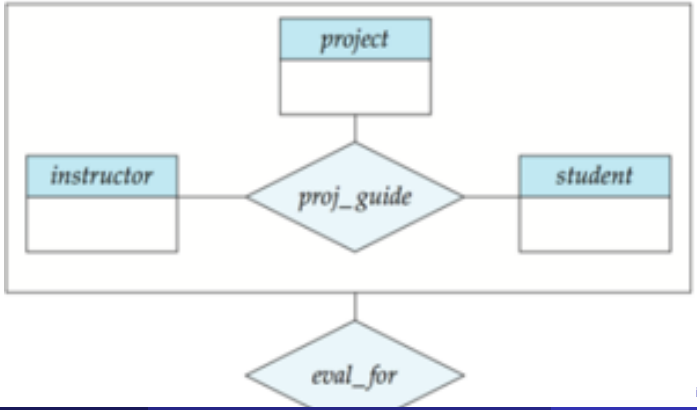

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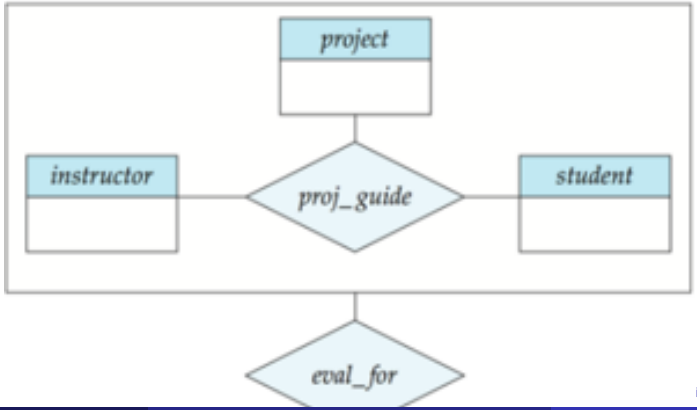

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Design Issues

Entities vs. Attributes

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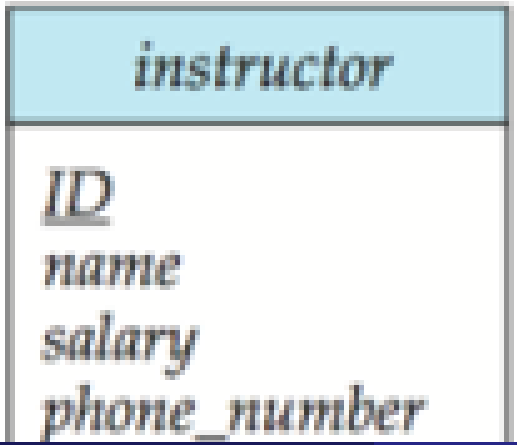

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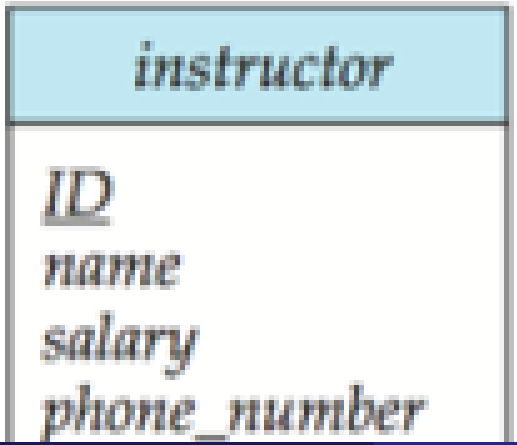
- Use of phone as an entity allows extra information about phone numbers (plus multiple phone numbers)



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  - For example, attribute `date` as attribute of `advisor` or as attribute of `student`

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  - Example: `proj_guide`

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  - d) add (e_i, c_i) to R_C

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- We can avoid creating an identifying attribute by making E, a weak entity set (described shortly) identified by the three relationship sets

ER Design Decisions

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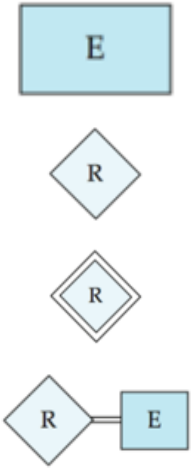
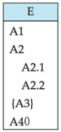
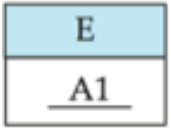
- The use of specialization/generalization - contributes to modularity in the design
- The use of aggregation - can treat the aggregate entity set as a single unit without concern for the details of its internal structure

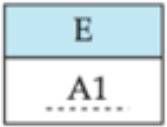
Symbols Used in ER Notation



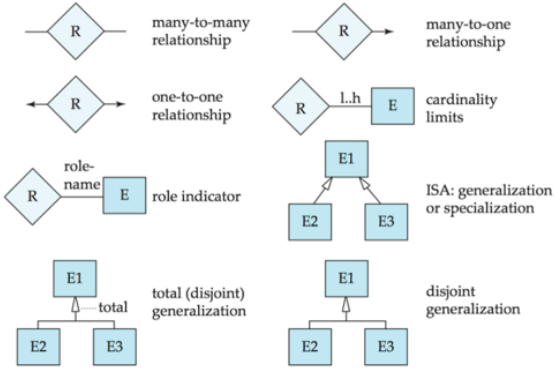
Symbols Used in ER Notation (Cont.)







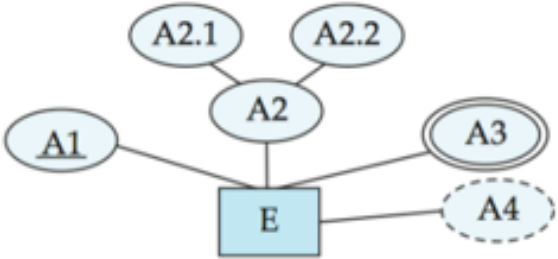
Symbols Used in ER Notation (2)



Symbols Used in ER Notation (3): Alternate

- Chen, IDE1FX, . . .

entity set E with
simple attribute A1,
composite attribute A2,
multivalued attribute A3,
derived attribute A4,
and primary key A1



weak entity set

generalization

total
generalization

Symbols Used in ER Notation (4): Alternates

Module 20
Partha Pratim Das
Objectives & Outline
ER Features
Non-binary Relationship
Specialization
Specialization as Schema
Generalization
Aggregation
Design Issues
Entities vs Attributes
Entities vs Relationship
Binary vs Non-Binary
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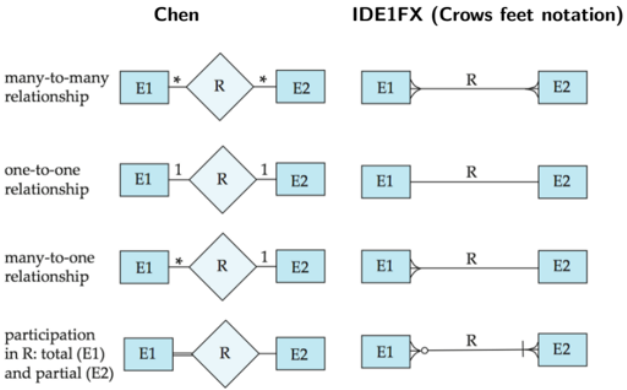
Module Summary



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Summary

- Discussed the extended features of ER Model

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- Discussed the extended features of ER Model
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